In [1]:
```
import pandas as pd
dataset=pd.read_csv("Social_Network.csv")
dataset
dataset=pd.get_dummies(dataset,drop_first=True)
dataset=dataset.drop(columns=["User ID"])
dataset
x=dataset[["Age","EstimatedSalary","Gender_Male"]]
y=dataset[["Purchased"]]
print(y)
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=1/8,random_state=52)
dataset[["Purchased"]].value_counts()
dataset.shape
```

```
     Purchased
0            0
1            0
2            0
3            0
4            0
..         ...
395          1
396          1
397          1
398          0
399          1

[400 rows x 1 columns]
```

Out[1]:

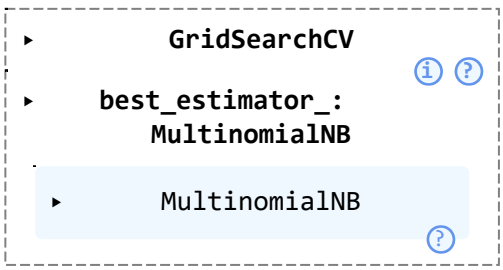
```
(400, 4)
```

In [2]:
```
from sklearn.naive_bayes import MultinomialNB
from sklearn.model_selection import GridSearchCV
param={ 'alpha':[1.0,3.0,5.0],
}
clf = GridSearchCV(MultinomialNB(),param,refit=True,verbose=3,n_jobs=-1)
clf.fit(x_train, y_train)
```

```
Fitting 5 folds for each of 3 candidates, totalling 15 fits
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversionWarning:
A column-vector y was passed when a 1d array was expected. Please change the shape of y to
(n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

Out[2]:



In [3]:
```
y_pred=clf.predict(x_test)
print(y_pred)
from sklearn.metrics import accuracy_score
```

```
accuracy=accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy * 100, "%")
```

```
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0
 0 0 1 0 0 1 1 0 1 0 0 0 0]
Accuracy: 70.0 %
```

In [4]:
```
from sklearn.metrics import confusion_matrix
r=confusion_matrix(y_pred,y_test)
print(r)
```

```
[[31 12]
 [ 3  4]]
```

In [5]:
```
from sklearn.metrics import classification_report
cr=classification_report(y_pred,y_test)
print(cr)
```

```
              precision    recall  f1-score   support

           0       0.91      0.72      0.81        43
           1       0.25      0.57      0.35         7

    accuracy                           0.70        50
   macro avg       0.58      0.65      0.58        50
weighted avg       0.82      0.70      0.74        50
```

In [ ]: